

OPERATIONAL AUTHORITY TOOLKIT

Open field templates for industrial architecture, integration and AI-enabled operations

Companion to “Operational Truth in the Age of AI”

BLACK + WHITE / PRINTABLE / EDITABLE / UNBRANDED

USE THESE AS WORKING TOOLS, NOT AS A CERTIFICATION SCHEME.

Choose the pages that help. Change the language. Remove fields. Add domain-specific detail. The objective is to make operational meaning, authority, execution, evidence and exception handling explicit enough that people and systems can reason about the same process.

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1. Authority Questions - Quick Scan

Use this before detailed interface design or automation. One answer may legitimately point to different systems or people.

Authority question	Working answer	Source / owner	Open?
Where is truth recorded? Authoritative representation of the object, state or decision.			[]
Who may command change? System, role or person allowed to issue or approve the instruction.			[]
Who executes? System, equipment or person that performs the action.			[]
Where does authoritative status come from? Source accepted as the resulting operational state.			[]
Who owns the evidence? Owner and location of proof that the decision/action/result occurred.			[]
Who owns the exception? Who decides what happens when the normal flow fails or sources disagree.			[]
Can a human override it? Who may intervene, under which conditions, and where that intervention is recorded.			[]
What remains unresolved? Decision still open; do not let implementation silently decide it.			[]

Tip: Different answers are not automatically a problem. Ungoverned or contradictory difference is.

2. Six-Verbs Authority Worksheet

A compact way to turn an ambiguous interface discussion into explicit governance questions.

Verb	Question	Answer / design decision
DEFINE	What object, state, instruction, decision, or evidence item are we actually talking about?	
AUTHORIZE	Who or what is allowed to issue or approve the decision?	
EXECUTE	Who or what performs the action?	
VALIDATE	Which system or role checks that the action is allowed in the current context?	
EVIDENCE	Where is proof of the decision, execution, result, and relevant context retained?	
OVERRIDE	Who may intervene when the normal flow fails, under what conditions, and how is that intervention recorded?	

Use the worksheet on one bounded object or decision at a time. “The MES” or “the ERP” is usually too broad a scope. “Production-order release”, “lot eligibility”, “quality disposition” or “machine readiness” is better.

3. State of the Answer

Separate what is known from what is proposed. This prevents workshop memory from becoming architecture by accident.

State	Use when	Evidence / reference
CURRENT FACT	Observed, recorded, verified, or otherwise established as the current state.	
CURRENT INFERENCE	Reasonable conclusion from available information, but not yet verified as fact.	
PROPOSED TARGET	Desired future state or design proposal. Not yet approved.	
UNRESOLVED DECISION	A decision is still open. Authority or target state is not yet agreed.	
APPROVED DECISION	Agreed and authorized target/decision. Implementation still needs separate evidence.	

Three guard rails:

- A proposal is not current state.
- A workshop statement is not approval.
- An approved target does not prove implementation.

Decision / statement to classify:

Selected state: ☐ Current fact ☐ Current inference ☐ Proposed target ☐ Unresolved decision ☐ Approved decision

4. Interface Authority Decision Record

Use one page per material decision, rule, state transition or interface responsibility.

Field	Working answer / decision
Decision / object	
Operational purpose / why it matters	
Current fact / inference	
Proposed / approved target	
Record of truth	
Command source / approver	
Execution owner	
Validation authority	
Authoritative status source	
Evidence owner + retention location	
Exception owner	
Human override: who / conditions / record	
Precedence if sources disagree	
Timing / sequencing assumptions	
Reconciliation rule	
Unresolved questions / dependencies	

Answer state: ☐ Current fact ☐ Current inference ☐ Proposed target ☐ Unresolved decision ☐ Approved decision

Decision owner / approver: _____ Date: _____ Evidence / link: _____

5. Record-of-Truth / Interface Authority Matrix

Use one row per bounded object, state, instruction or decision. The matrix is a forcing function, not the end product.

Object / decision	Record of truth	Command source	Execution owner	Validate	Status source	Evidence owner	Exception owner	Human override	Answer state	Notes / unresolved

Answer state: Current fact | Current inference | Proposed target | Unresolved decision | Approved decision

Check for: duplicated selection rules; hidden precedence; timing assumptions; unowned reconciliation; unowned exceptions; human intervention with no evidence trail.

6. End-to-End Operational Model Canvas

The matrix is not the destination. Use this canvas to assemble the governed operational picture that people and machines need to share.

ACTORS Systems, roles, people, equipment. Who participates?	OBJECTS + STATES Orders, lots, recipes, samples, machine states, decisions, evidence. What exists?
MEANING What do the key objects, terms and states mean in this operational context?	AUTHORITY Who may decide, command, approve, validate, release, reject or override?
ACTIONS + EXECUTION What happens, in what sequence, and who or what performs each action?	EVIDENCE What proves the decision, action, result and relevant context? Where is it retained?
EXCEPTIONS + OVERRIDES What happens when the happy path breaks? Who owns recovery and intervention?	RELATIONSHIPS + HANDOFFS How do actors, objects, meanings and authority cross system/process boundaries?

Unresolved decisions / assumptions that must not disappear into implementation:

The objective is not one system owning everything. The objective is deliberate, visible and traceable boundaries.

7. Pre-Automation Authority Check

Run this before delegating recommendations, routing, approvals, triggers or actions to software or AI.

Check	Yes / No / N/A	Gap / action
What does the relevant state / object / instruction mean?	☐ Yes ☐ No ☐ N/A	
Which source is authoritative for this specific purpose?	☐ Yes ☐ No ☐ N/A	
Who may make or approve the decision?	☐ Yes ☐ No ☐ N/A	
What may the software recommend, change, trigger or execute?	☐ Yes ☐ No ☐ N/A	
What must remain human-controlled?	☐ Yes ☐ No ☐ N/A	
What proves what happened?	☐ Yes ☐ No ☐ N/A	
Who owns the exception when normal flow fails?	☐ Yes ☐ No ☐ N/A	
Who may override the normal flow?	☐ Yes ☐ No ☐ N/A	
Where is that override recorded?	☐ Yes ☐ No ☐ N/A	
What happens when two valid sources disagree?	☐ Yes ☐ No ☐ N/A	
Is precedence explicit?	☐ Yes ☐ No ☐ N/A	
Are timing and reconciliation rules explicit?	☐ Yes ☐ No ☐ N/A	
Can a new team member reconstruct why this authority was assigned?	☐ Yes ☐ No ☐ N/A	

Decision to automate / delegate:

Go / no-go conclusion: ☐ Proceed ☐ Proceed with controls ☐ Hold - authority unresolved

8. Reuse Note + License

These templates are deliberately method-light and unbranded.

The tools in this pack are intended to make operational meaning and authority visible before they disappear into implementation. They are not an industry standard, an ATS methodology, a certification model, or a universal requirement.

Use the parts that help. Rename fields to fit the domain. Add regulatory or site-specific controls where needed. Replace the matrix entirely if an existing decision record or architecture practice captures the same information better.

The end goal is not a completed worksheet. It is a coherent end-to-end operational model in which actors, objects, states, relationships, meanings, authorities, actions, evidence and exceptions are explicit enough to be understood consistently across the process.

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Companion article: "Operational Truth in the Age of AI"